

1. K-means Clustering

Definition:

K-means clustering is an **unsupervised machine-learning algorithm** that divides data into **K groups (clusters)** based on similarity.

How it works:

Data points



Choose K



Place K centroids



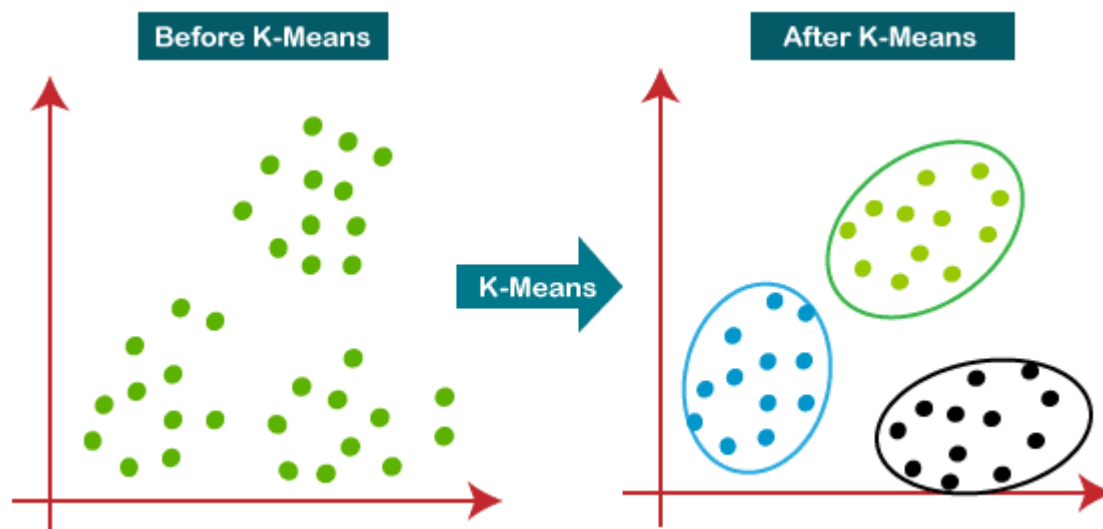
Assign each point to nearest centroid



Move centroids to cluster centers



Repeat until clusters stabilize



Usage:

- Customer segmentation
- Grouping similar patients
- Image segmentation
- Market research
- Finding patterns in datasets

Key point:

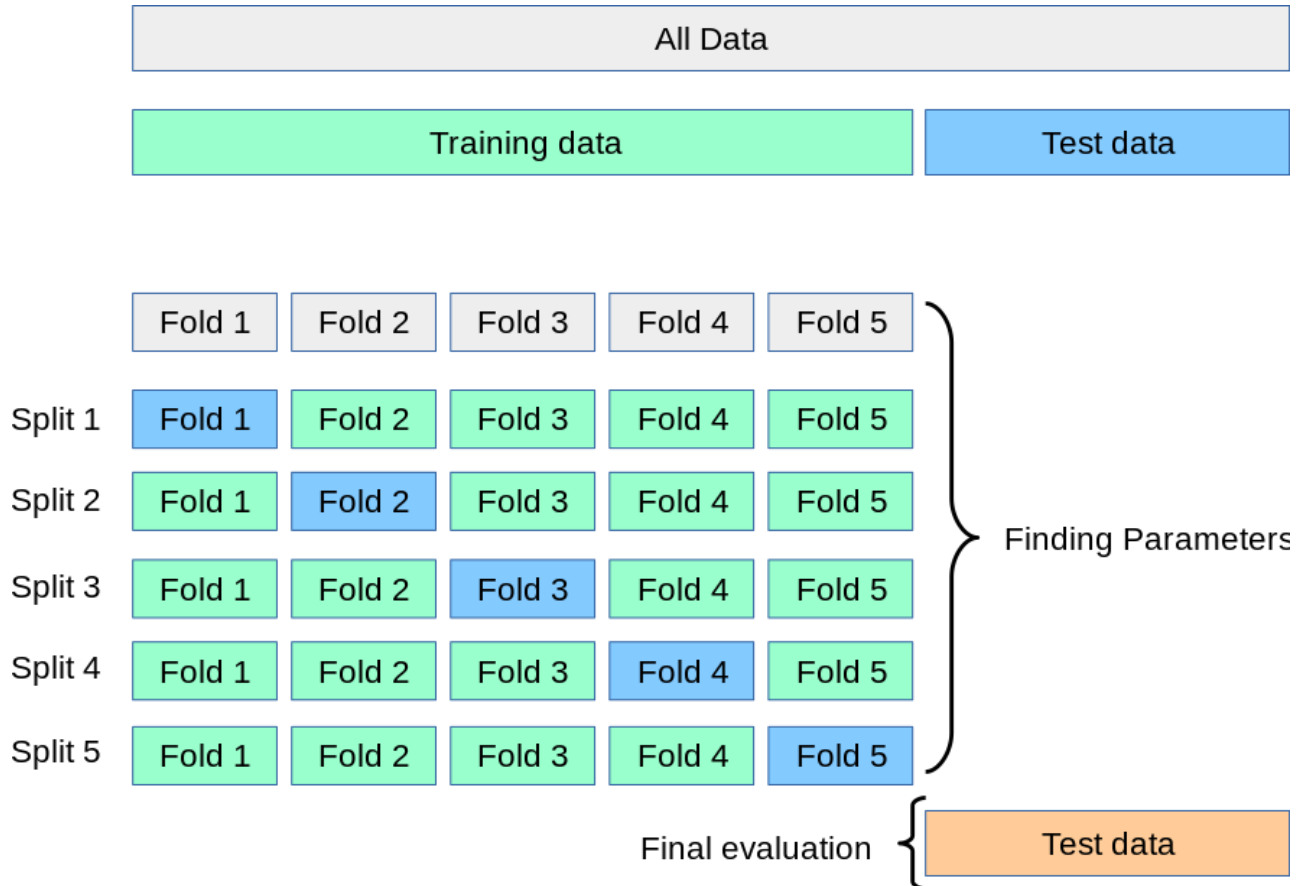
K-means = group similar observations without using predefined labels.

2. Cross-validation

Definition: Cross-validation is a technique used to evaluate how well a machine-learning model will perform on **new, unseen data**.

The most common method is **K-fold cross-validation**.

Graphic example: For **5-fold cross-validation**:



The model is trained and tested **5 times**.

Accuracy:

Round 1 → 82%

Round 2 → 86%

Round 3 → 84%

Round 4 → 88%

Round 5 → 85%

Average = 85%

Usage:

- Comparing machine-learning models
- Selecting the best model
- Detecting overfitting
- Choosing hyperparameters
- Getting a more reliable estimate of model performance

Key point:

Cross-validation = test the model several times using different portions of the data.

3. Confusion Matrix

Definition:

A confusion matrix is a table used to evaluate a **classification model** by comparing actual classes with predicted classes.

Binary classification

Where:

- **TP (True Positive)**: Correctly predicted positive
- **TN (True Negative)**: Correctly predicted negative
- **FP (False Positive)**: Predicted positive but actually negative
- **FN (False Negative)**: Predicted negative but actually positive

		Predicted Values		
		Predicted Positive (1)	Predicted Negative (0)	
Actual Values	Positive (1)	TP = 603	FN = 47	Actual Positive Total = 650
	Negative (0)	FP = 29	TN = 321	Actual Negative Total = 350
		Predicted Positive Total = 632	Predicted Negative Total = 368	n = 1000

Visual interpretation:

Prediction
Correct Wrong

Positive	TP	FN
Negative	TN	FP

Usage:

- Classification model evaluation
- Medical diagnosis
- Fraud detection
- Spam detection
- Disease prediction
- Survival prediction

Key point:

Confusion matrix = shows exactly where a classification model is correct and where it makes mistakes.

4. Sensitivity and Specificity

Sensitivity

Definition:

Sensitivity measures how well a model identifies **actual positive cases**.

Sensitivity =

True Positives

True Positives + False Negatives

Think:

Of all people who actually have the disease, how many did the model detect?

Example:

100 sick patients

↓

90 correctly detected

10 missed

Sensitivity = 90%

Specificity

Definition:

Specificity measures how well a model identifies **actual negative cases**.

Specificity =

True Negatives

True Negatives + False Positives

Think:

Of all people who are actually healthy, how many did the model correctly identify as healthy?

Example:

100 healthy people

↓

95 correctly identified

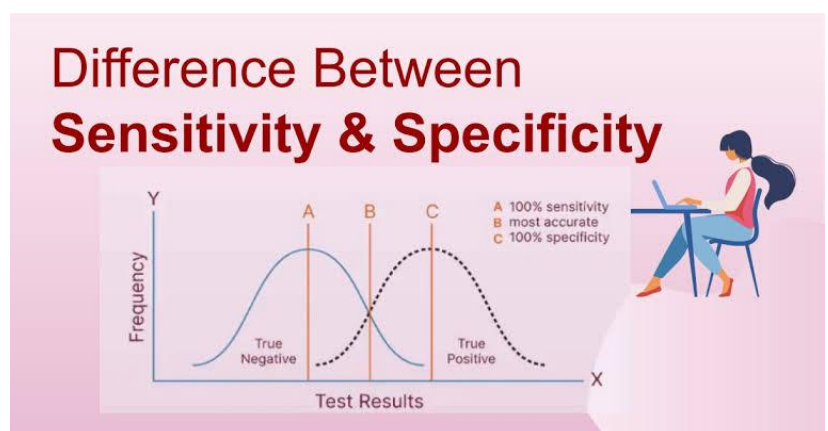
5 incorrectly classified as sick

Specificity = 95%

Easy memory trick:

Sensitivity → Sick detected

Specificity → Healthy detected



5. Bias–Variance

Definition:

Bias and variance describe two major sources of prediction error in machine-learning models.

Bias

Bias = model is too simple.

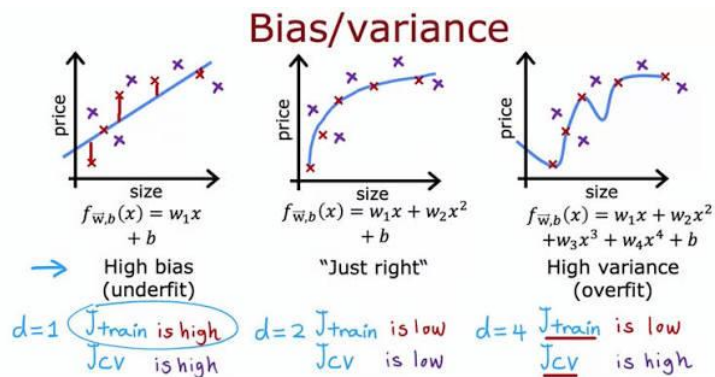
The model makes strong assumptions and misses important patterns.

Example:

Actual data:

• •
• •
•
• •
• •

High-bias model:



The model is **underfitting**.

Variance

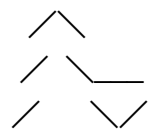
Variance = model is too sensitive to the training data.

It learns noise instead of the general pattern.

Actual data:

• •
• •
•
• •

High-variance model:



The model is **overfitting**.

The goal

High Bias

Underfitting

Too simple

High Variance

Overfitting

Too complex

↓
Balanced model



Good generalization

Usage:

- Choosing model complexity
- Diagnosing underfitting
- Diagnosing overfitting
- Improving model performance

Key point:

Bias = too simple. Variance = too sensitive.

$$Var(X) = E(X^2) - [E(X)]^2$$

$$Var(X) = (1.4)^2 = 1.96$$

σ

σ

$$\mu - \sigma + \sigma Var(X) \approx 1.96$$

Give feedback

6. Entropy

Definition:

Entropy measures **impurity or uncertainty** in a dataset.

It is commonly used in **decision trees** to decide the best split.

Example 1: Pure group

10 passengers

Survived: 10

Died: 0



Entropy = 0

There is **no uncertainty**.

Example 2: Mixed group

10 passengers

Survived: 5

Died: 5

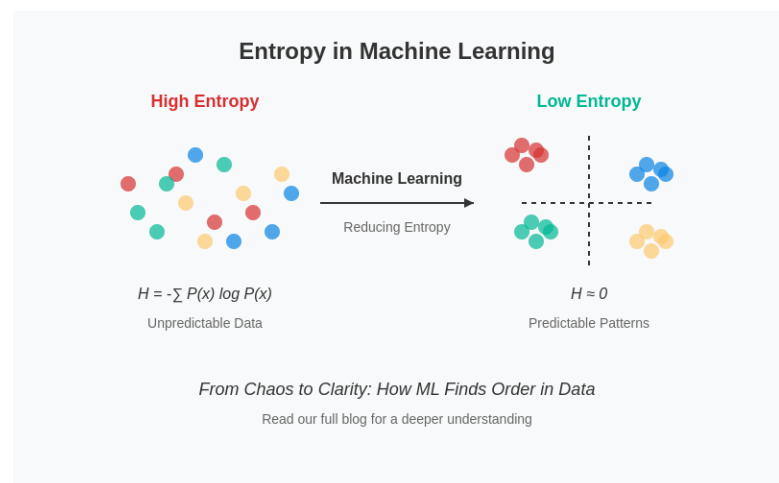


High entropy

There is **high uncertainty**.

Pure group

Mixed group





Low entropy



High entropy

Usage:

- Decision trees
- Selecting the best data split
- Measuring class impurity
- Classification problems

Key point:

Low entropy = more pure. High entropy = more mixed/uncertain.

7. Linear Regression

Definition:

Linear regression is a **supervised machine-learning/statistical method** used to predict a continuous numerical value from one predictor.

The basic equation is:

$$\hat{y} = b_0 + b_1x$$

Where:

- $\hat{y}$ = predicted value
- b_0 = intercept
- b_1 = slope
- x = predictor

$$\hat{y} = b_0 + b_1x$$
$$\hat{y} = 8.49 - 0.54x$$

$R^2 = 0.72$ · b_0 = intercept · b_1 = slope · least squares minimizes squared vertical residual gaps.

Focus

DataLinePredictionsResiduals

DataLinePredictionsResiduals

Drag a point to refit the line.

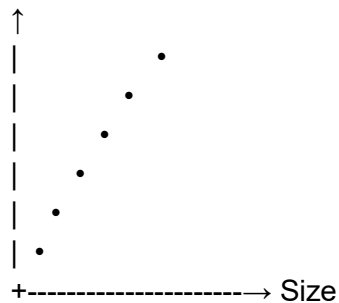
246810246810

Give feedback

Example

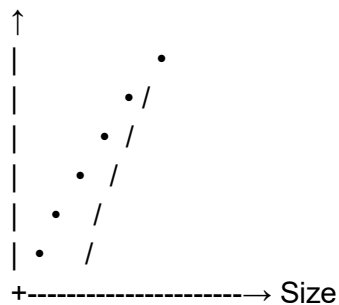
Suppose we want to predict **house price from house size**.

Price



The regression line finds the best-fitting straight line:

Price



Example:

House size = 2,000 sq ft

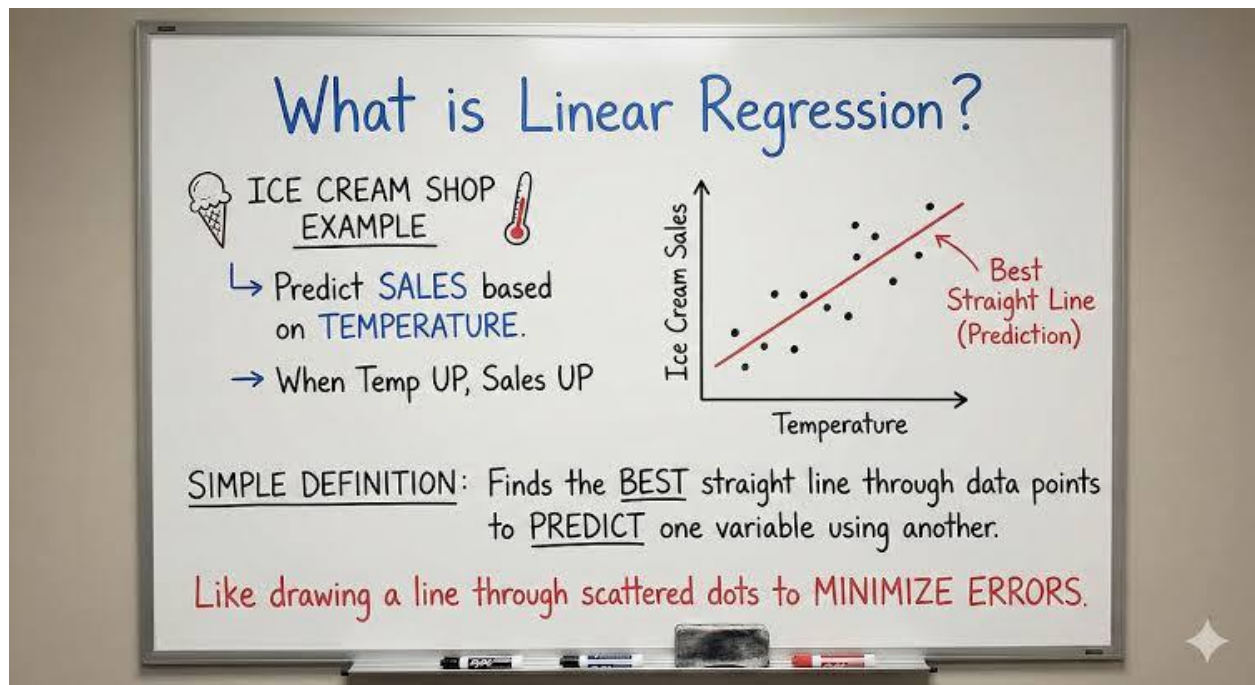
Predicted price = \$250,000

Usage:

- Predicting prices
- Predicting yield
- Predicting income
- Predicting temperature
- Studying relationships between variables

Key point:

Linear regression = predict a continuous outcome using one predictor.



8. Multiple Regression

Definition:

Multiple regression predicts one continuous outcome using **two or more predictor variables**.

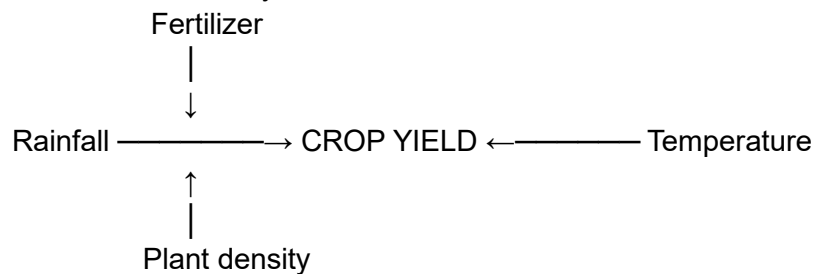
The general equation is:

$$\hat{y} = b_0 + b_1x_1 + b_2x_2 + b_3x_3 + \dots$$

Example

Predicting crop yield using:

- Fertilizer
- Rainfall
- Temperature
- Plant density



Instead of:

Yield $\leftarrow$ Fertilizer

we use:

Yield $\leftarrow$ Fertilizer

+ Rainfall

+ Temperature

+ Plant density

Example equation

Yield = 1.5

+ 0.8(Fertilizer)

+ 0.3(Rainfall)

- 0.2(Temperature)

Usage:

- Predicting crop yield
- Predicting house prices
- Economic forecasting
- Agricultural experiments
- Identifying effects of multiple factors

Key difference:

Method	Number of predictors	Example
Linear regression	1	Yield $\leftarrow$ Fertilizer
Multiple regression	2 or more	Yield $\leftarrow$ Fertilizer + Rainfall + Temperature

Easy memory:

Linear regression = one $X \rightarrow Y$

Multiple regression = many $Xs \rightarrow$ one

